Chapter 4
Estimating and Reducing Labor Costs (Process Improvement by Line Balancing)

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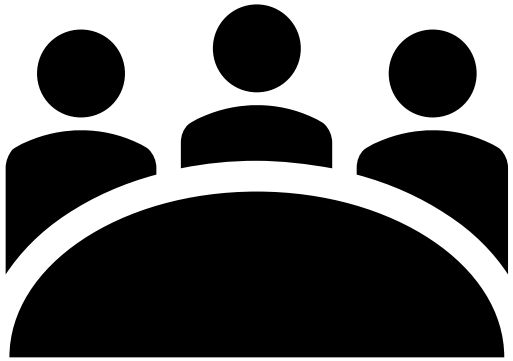
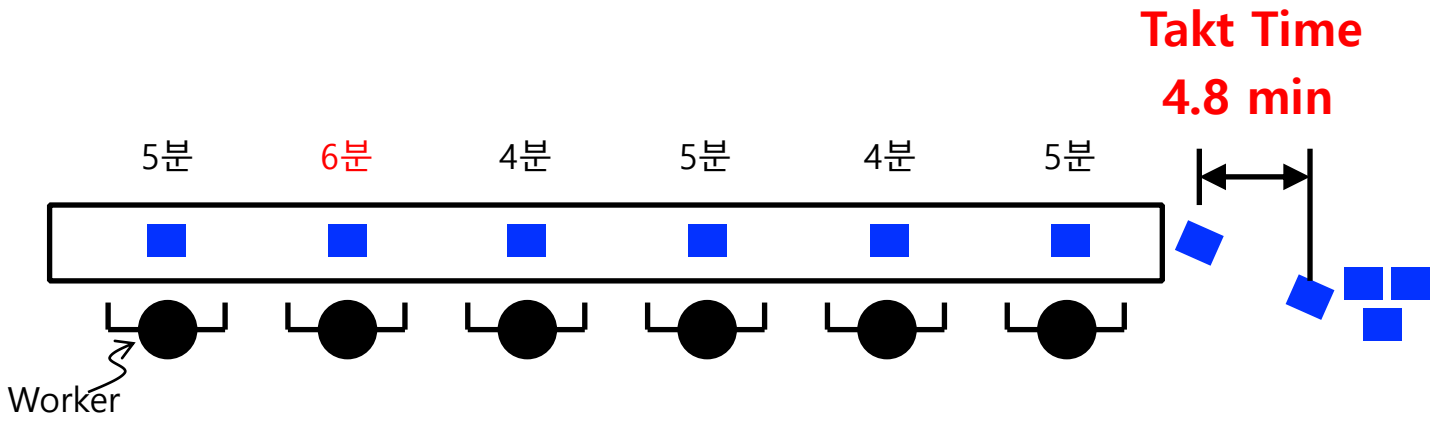
Collaborative Clarification

- Students raise questions about parts they found difficult
- Any students may answer and one additional student may add comments
- The professor clarifies both the questions and answers if necessary
- Students who raised questions and answered write down what they did (A Q&A sheet)



Sample Assembly Line

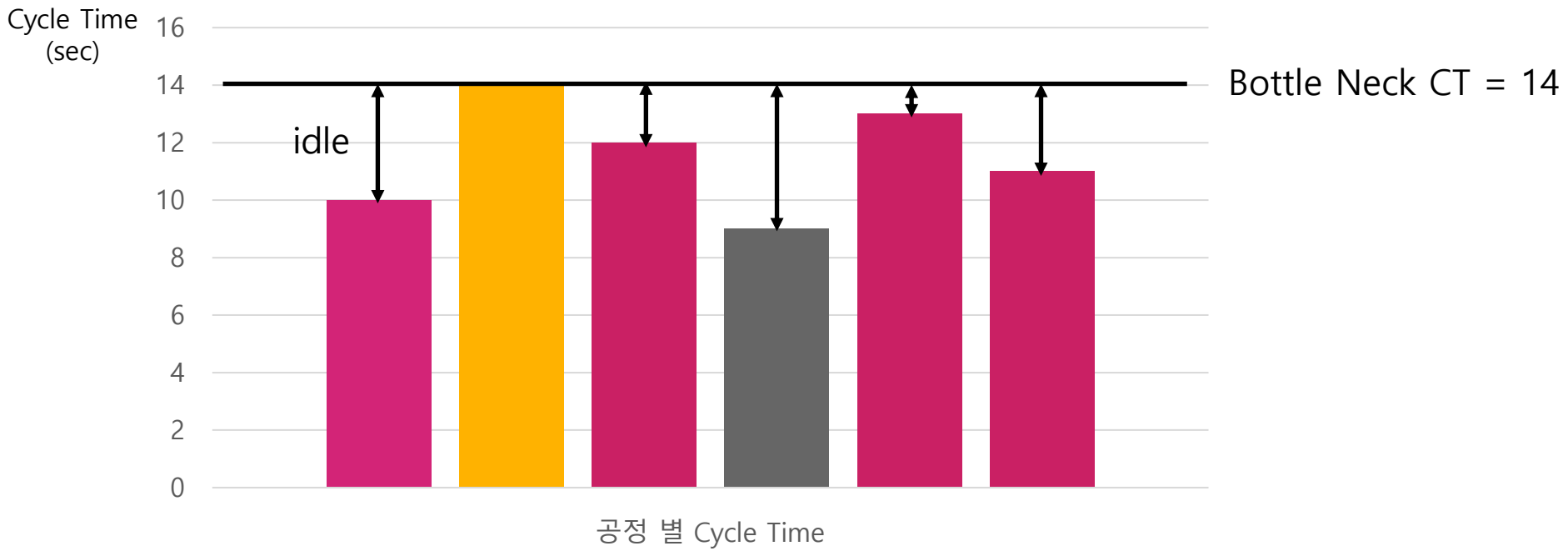
- Takt Time = $\dfrac{\text{Working hours a day}}{\text{Required production amount a day}}$
  - ex, 8hrX60min/100pairs=4.8min/pair, a pair should be out every 4.8min
- The assembly line is designed so that its cycle time is approximately equal to the takt time
  - The cycle time (CT) of the assembly line below is the same as the bottleneck cycle time



CT of Assembly Line
= CT of Bottle Neck
= 6 min

Line Balancing (LOB)

A metric that indicates how busy each process is without idle time



$$LOB = \frac{\text{Sum of each station's cycle time}}{\text{BottleNeck CT (Line CT)} \times \text{\# of stations}} = \frac{10+14+12+9+13+11}{14 \times 6} = \frac{69}{84} = 82\%$$

Applied Learning

- Distribute a short case and its accompanying podcast if necessary
  - POM 4 (Applied Learning) Labor Costs and Production Efficiency
- Students would share their thoughts with peers
- Students should write down their reflections (A reflection sheet)

Debating Statement

- In high-volume assembly, breaking work into very small specialized tasks does not always improve efficiency. Because tasks cannot be divided endlessly, parallel work cells can sometimes perform better than a highly specialized assembly line
  - Splitting work into tiny, repeated steps isn't always efficient. Sometimes it is better to have separate teams build the whole product in parallel, rather than using one long assembly line where tasks are sliced too thin.
  - **The "Limit" Focus** "There is a limit to how much you can divide a job. If you chop tasks into pieces that are too small, you waste time. It is often faster to use 'parallel cells' where one person does multiple steps."
  - **The "Assembly Line" Critique** "The traditional assembly line has a flaw: you can't divide work endlessly. Often, it is smarter to have five people each build a complete product (Parallel) than to have a line of five people where each does 20% of the work."
- **The "YES" Argument (Focus: Efficiency & Motivation)**
  - *Stop slicing the cake into crumbs; let people eat a slice.*
- **The "NO" Argument (Focus: Speed & Cost)**
  - *Specialization is the engine of the industrial revolution.*

Debating Statement

- The Toyota Production System tries to eliminate all idle time as waste, but this can be risky. It may improve short-term productivity while missing opportunities to uncover and solve root causes of problems
  - Toyota's method of eliminating all waiting time can backfire. If workers are always 100% busy, they never have the time to stop, think, and fix the real reason problems are happening.
  - **The "Hamster Wheel" Analogy** "Trying to kill every second of idle time is risky. It creates a 'hamster wheel' where everyone runs fast to boost numbers today, but nobody steps off the wheel to fix the machine for tomorrow."
  - **The "Slack" Focus** "While Toyota considers idle time to be waste, removing it entirely is dangerous. Without a little bit of 'slack' or free time in the system, teams lose the ability to investigate and solve the root causes of errors."
- **The "YES" Argument (Focus: Resilience & Improvement)**
  - *You need breathing room to think.*
- **The "NO" Argument (Focus: Discipline & Visibility)**
  - *Idleness is the enemy of quality.*

Thank You

Matching supply and demand

- In Chapter 3
  - We learned utilization, implied utilization, capacity, and bottleneck related to the process and resource within the process
  - Maximizing resource utilization only is not necessary
  - Our objective is to match supply with demand in order to create value (= to make profits) ➔ implied utilization
- In this chapter
  - Let's learn how to improve the process, when the excess capacity can be reduced, or can be added to the bottleneck
  - The concept of line balancing is needed

Revisited

Process Step	Max # of flow units processed	A given unit of time (hr)	Max flow rate (# of flow units/hr)
Parboil	16	1	16
Burn	8	1	8
Cut	20	1	20
Spread Cheese	10	1	12
Wrap	10	1	10

Process Capacity

Process Step	Max # of flow units processed	A given unit of time (hr)	Max flow rate (# of flow units/hr)	Flow time per unit (min)
Parboil	16	1	16	1/16 *60 = 3.75
Burn	8	1	8	7.5
Cut	20	1	20	3
Spread Cheese	10	1	10	6
Wrap	10	1	10	6

26.25

Labor Cost per Unit

- Basic Assumption
  - Assume hourly rate = 9000₩/hr = 150₩/min
  - Each process step has a worker
- Labor cost for making a unit
  - Flow time = 26.25 min/unit
  - Labor cost per unit = 26.25 * 150 = 3937.5₩/unit
- Is this right?

Assume Demand=40/day

- Let's calculate the labor cost per unit, given daily demand < process capacity (=64/day)
- Labor cost per unit
  - The workers should be paid daily regardless of the number of units produced

$$\frac{5 \text{ workers} \times 8 \text{ hours/day} \times 9000₩/\text{hour}}{\cancel{50}\ 40 \text{ units/day}} = \cancel{7200}\ 9000₩/\text{unit}$$

Idle Time Calculation for Demand=40/day

- Process flow rate per hour = min {Process Capacity, Demand rate} = Min {8, 40/8} = 5

Process Step	Max # of flow units processed	A given unit of time (hr)	Max flow rate (# of flow units/hr)	Max flow rate – Process flow rate	Flow time per unit (min)	Idle time(min)/hr
Parboil	16	1	16	11	1/16 *60 = 3.75	11X3.75
Burn	8	1	**8**	**3**	**7.5**	**3X7.5**
Cut	20	1	20	15	3	15X3
Spread Cheese	10	1	10	5	6	5X6
Wrap	10	1	10	5	6	5X6

Assume Demand=120/day

- Let's calculate the labor cost per unit, given daily demand > max capacity (8/hr=64/day)
- Cost of direct labor

$$\frac{5 \text{ workers} \times 8 \text{ hours/day} \times 9000₩/\text{hour}}{64 \text{ units/day}} = 5625₩/\text{unit}$$

- Why is this smaller than previous one?

Idle Time Calculation for Demand=120/day

- Process flow rate per hour = min {Process Capacity, Demand rate} = Min {8, 15} = 8

Process Step	Max # of flow units processed	A given unit of time (hr)	Max flow rate (# of flow units/hr)	Max flow rate – Process flow rate	Flow time per unit (min)	Idle time(min)/hr
Parboil	16	1	16	8	1/16 *60 = 3.75	8X3.75
Burn	8	1	**8**	**0**	**7.5**	**0X7.5**
Cut	20	1	20	12	3	12X3
Spread Cheese	10	1	10	2	6	2X6
Wrap	10	1	10	2	6	2X6

Could you make more units by improving the process?

How to Break up a Process Step

- The process is constrained by the Burn
- Let Burn borrow 1min from Cut
  - That is, reallocate work items to the underutilized process
  - For example, the Burn consists of 1 min for setup, 3 min for actual burning, 1 min for cleaning, then the cleaning work can be moved to the Cut
- Other ways?
  - Increase the efficiency (ex, by improving the burning process)
  - Increase the capacity (ex, add one more labor, or reallocating labor from different process if any)

Process Improvement

- Allocate 1 minute from Cut to Burn
- Modify the following table to see how many flow units flow a day

Process Step	Max # of flow units processed	A given unit of time (hr)	Max flow rate (# of flow units/hr)	Max flow rate – Process flow rate	Flow time per unit (min)	Idle time(min)/hr
Parboil	16	1	16	8	1/16 *60 = 3.75	8X3.75
Burn	8	1	**8**	**0**	~~**7.5**~~ 6.5	**0X7.5**
Cut	20	1	20	12	3 ~~X~~ 4	12X3
Spread Cheese	10	1	10	2	6	2X6
Wrap	10	1	10	2	6	2X6

What if Explosive Increase in Demand

- Discuss how to cope with explosive increase in demand in order not to lose opportunity
  - In case demand increases from 120 to 600

Breakout

Note that the team leader should write down the key idea with his(her) name from each team member.